

Amjad S Althabteh

[LinkedIn](#) | [Portfolio](#) | [✉ Amjadjamalshadi@gmail.com](mailto:Amjadjamalshadi@gmail.com) | [Github](#) | 734-657-3714

Summary

C++ graphics and engine developer specializing in real-time rendering and performance-critical systems. Built modular OpenGL engines, custom allocators, and GPU-optimized pipelines. Experienced in Unreal Engine 5 gameplay systems and low-level debugging, with strong focus on memory efficiency and real-time performance.

Education

B.S in Computer Science	GPA: 3.65/4.00	Wayne State University	Detroit	2026
-------------------------	----------------	------------------------	---------	------

Experience

Graphics Engineer, Co Founder	Immersalab Graphics	Detroit, MI	Sep 2025 - Dec 2025
--------------------------------------	----------------------------	-------------	---------------------

- Architected a real-time C++20/OpenGL rendering engine with GPU-driven pipelines and performance-focused design.
- Built internal UI and JSON-based asset pipelines for flexible content creation.
- Implemented data-driven rendering optimizations for performance balancing.
- Added GPU profiling and frame-timing tools for debugging and optimization.

Unreal C++ Dev	Indie Game AAA	Ann Arbor, MI	Apr 2025 - Aug 2025
-----------------------	-----------------------	---------------	---------------------

- Developed gameplay systems in C++20 and visual scripting, integrating animation and narrative logic.
- Created and rigged 3D environments and characters.
- Produced a 90+ second real-time cinematic using MetaHuman, Control Rig, and Sequencer.
- Built animation control systems with C++ and Animation Blueprints for smooth blending.

[no_one_answers_slice](#)

Software Development Intern	Rare Munchies	Ann Arbor, MI	Sep 2023 - Apr 2024
------------------------------------	----------------------	---------------	---------------------

- Integrated UI components using **TypeScript**, improving page transitions and interaction flow.
- Identified and resolved **cross-browser compatibility issues**, reducing debugging time and unblocking delivery
- Optimized client-side rendering logic and state management patterns to reduce unexpected UI re-renders.

Projects

3D Space Simulation Engine – C++/OpenGL	space-simulations
--	-----------------------------------

- Built a custom 3D space simulation engine in C++17 using OpenGL 4.x, GLAD, and SFML.
- Developed modular engine architecture (rendering, physics, math, entities, and universe systems).
- Implemented custom math library (Vec3, Mat4) and camera system with projection/view matrices.

HFT OrderBook Engine C++ 20

- Built a modular price-time FIFO matching engine with bid/ask depth tracking and partial fill execution logic.
- Designed low-latency infrastructure using cache-aligned data structures and modern C++20 features to minimize branch misprediction and memory overhead.

Glint 3D – C++ Automated 3D Rendering Engine

[glint-rendering](#)

- Built a C++17/OpenGL rendering engine with optimized GPU workload and lighting.
- Cross-platform architecture with runtime scene analysis for stable performance.
- Implemented custom C++20 memory pool allocator, reducing heap allocations and improving runtime stability under load.

C++ Runtime Stack Trace & Diagnostics Engine (Traceforge)

- Built a C++23 debugging system for automated stack-trace parsing.
- Automated regression test generation with intelligent fix suggestions.
- Added memory tracking utilities to detect leaks and invalid allocations.

Skill

- **Programming Languages:** C++ (C++11–C++23), C, C#, Python, TypeScript, Java
- **Graphics & Engines:** OpenGL • Unreal Engine 4 - 5.7 • Unity • Godot • NX Design
- **Course Work:** Computer Graphics, Game Programming, **Software Engineering**, **Operating Systems**, Linear Algebra
- **Specialties:** Gameplay Systems, VR/AR Development, Simulation, Tools & Plugins, Technical Art